HAND-GESTURE DETECTION AND ASL TRANSLATION APPLICATION: INTEGRATION OF IMAGE PROCESSING AND SENSORS

ỨNG DỤNG PHÁT HIỆN CỬ CHỈ TAY VÀ DỊCH ASL: TÍCH HỢP XỬ LÝ HÌNH ẢNH VÀ CẢM BIẾN

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Abstract:

The language barrier presents a significant challenge to effective communication and relationship-building, particularly for individuals who are unable to communicate verbally, such as those with hearing impairments or language disorders who rely on sign language. Acquiring a new language demands time and effort, which may not be readily available in our fast-paced society. Various initiatives have been undertaken to bridge the communication gap between sign language users and non-users, including the use of data gloves, sensors, and machine learning for image processing. This study proposes a new hybrid approach that combines computational image analysis and sensor-focused techniques to facilitate a better understanding of sign language for non-users. The objective is to promote accessibility for individuals who primarily use sign language for communication.

Keywords:

Deep Neural Network, Hand Recognition, Sensor-based sign translator, Vision-based sign translator

Tóm tắt:

Rào cản ngôn ngữ là một thách thức đáng kể đối với việc giao tiếp hiệu quả và xây dựng mối quan hệ, đặc biệt đối với những cá nhân không thể giao tiếp bằng lời nói, chẳng hạn như những người khiếm thính hoặc rối loạn ngôn ngữ phải dựa vào ngôn ngữ ký hiệu. Việc tiếp thu một ngôn ngữ mới đòi hỏi thời gian và công sức, những điều này có thể không có sẵn trong xã hội phát triển nhanh chóng của chúng ta. Nhiều sáng kiến khác nhau đã được thực hiện nhằm thu hẹp khoảng cách giao tiếp giữa người sử dụng ngôn ngữ ký hiệu và người không sử dụng, bao gồm việc sử dụng găng tay cảm biến và các phần mềm xử lý hình ảnh. Nghiên cứu này đề xuất một phương pháp mới kết hợp phân tích hình ảnh tính toán và các kỹ thuật tập trung sử dụng cảm biến để tạo điều kiện hiểu rõ hơn về ngôn ngữ ký hiệu cho những người không quen thuộc, thúc đẩy khả năng tiếp cận cho những cá nhân chủ yếu sử dụng ngôn ngữ ký hiệu để giao tiếp.

Từ khóa:

Mạng neuron sâu, Nhận dạng bàn tay, Trình dịch dấu hiệu tay dựa trên cảm biến, Trình dịch dấu hiệu tay dựa trên hình ảnh

1. INTRODUCTION

According to the World Health Organization (WHO), hearing impairment impacts over 430 million people worldwide[1]. In stark contrast, the report also demonstrated that supportive products for people with hearing loss are distributed and benefited in middle to low-income countries by only 3% (2019). Much research has been conducted in an attempt

to bridge the gap between sign language users and non-users, the most notable solutions are between the utilization of data gloves[2], [3], [4] and sensors, or the quickly-evolving subfield of machine learning that is image processing. Prosthetics and wearable devices are used in conjunction with flexible sensors that measure the bending degrees of our fingers and accelerometers to measure different hand positions and return readings based on the data collected. Alternatively, image processing is an ongoing and rapidly developing field, producing many promising results with high-accuracy predictions and real-time translation of hand signs, using Hand Recognition (HR) models built upon Convolutional Neural Networks (CNN)[5], [6] and frameworks such as Tensorflow.

Although there are proven benefits, there are also criticisms regarding both of these methods. The glove-based solution, although effectively portable, poses many challenges for the user due to its inherent restrictive design, preventing the user from performing complex signs and creating discomfort for the person[7], while vision-based methods, while well-regarded for their accuracy, are criticized for their inability to deal with changing environmental conditions like lighting or non-uniform background, as well as their non-portability and having the user move in a restricted space to be in the frame of detection. The deployment of real-time image processing techniques into a portable device also creates its own set of challenges due to the hardware limitations in smaller machines, contrasting the demands for higher computational power for such applications.

This study aims to examine a new approach, a hybrid between the heavy computational image analysis and the sensor-focused idea of the data gloves. The focus here is to embrace the most effective parts of each method while using one to improve where the other is ineffective. Moreover, research on a lower-cost, portable device capable of real-time sign language translation while employing a hybrid of computer vision and sensor-based approaches remains limited.

This project's objective is to develop a low-cost and relatively effective solution that can aid communication for the hearing-impaired and create an accessible gateway for the general public to familiarize themselves with sign languages. Utilizing sensors and servo motors with microcontrollers to reinforce the effectiveness of a machine algorithm that can translate sign language, specifically letters in the American Sign Language (ASL) alphabet, into English via real-time camera inputs, disabling many weaknesses presented by these applications when operated individually

2. METHODOLOGY

2.1. Proposed System

The general concept of this sign language translator is to use image processing and sensors in combination to detect the hands, their orientation, movement, and environmental factors that can affect the translation process when performing the sign language, then the hand gesture data

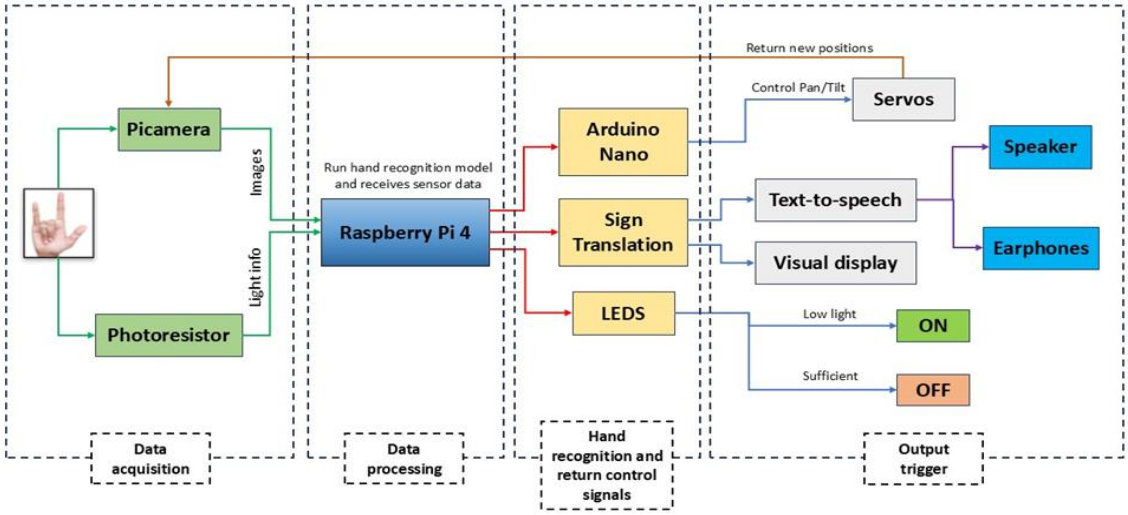

Figure 1: The system architecture

will be stored and processed by a neural network algorithm within the Raspberry Pi computer. This database stores each sign's characteristic finger positions as key point coordinates, associated with different articulations. Providing the position of each finger to detect which gesture the hand is making. Once a match is found, the program can trigger the output, which could be either a text of the translation on a screen or a text-to-speech corresponding audio associated with the recognized sign in the database. At the same time, a photoresistor sensor will be implemented to detect the light quality in the room, detecting when the ambient brightness is on lower levels and activating an array of LEDs on the device, illuminating the subject and enhancing detection visibility. The camera itself is attached to a Pan-and-Tilt setup that involves two servos controlled by the detection program through the use of an Arduino microcontroller, its job is to track the orientation of the subject's hands and use the two attached servos to position the detected hands in a way that the user is always within the frame of detection, addressing the issue of the restricted space. The established framework of Mediapipe will be applied to the dataset preparation and hand-tracking applications. The entire development pipeline will include data collection, preprocessing, model training, classification, sensor integration, and finally system implementation. The proposed system for this translator is shown in Figure 1.

The sign language translation prototype device is designed to be multi-functional, user-friendly, portable, and unobtrusive while capturing the necessary hand movements for accurate sign recognition. The prototype is composed of small modules attached to a board, with a plastic casing for the main processing unit and pan-tilt components for hand-tracking applications. The sensors are mounted onto the pan and tilt component itself to record the environmental changes, while hand signs are detected in real time. These sensors will be connected to a

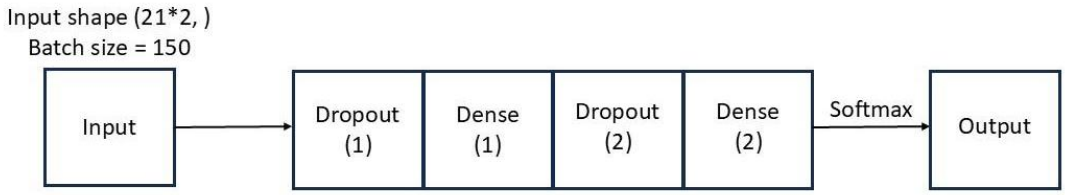

Figure 2: The architecture of the neural network model

microcontroller (Arduino Nano) powered by the main computer (Raspberry Pi 4 Model B), which also drives the servo motors and processes the camera inputs. The utilization of an Arduino control allows for more processing power of the Raspberry Pi to be focused on the task of detecting and translating the hand signs. The audio output will be positioned within the system and can be accessed via a 3.5mm jack through a speaker or a headphone, while the visual output can be displayed either on a monitor connected via HDMI or the embedded OLED screen on the device.

2.2. Model architecture

The design of the hand-recognition prototype device can be separated into two parts, each handling different aspects of the detection and translation process, one compensating for the drawbacks of the other. The software includes the hand detection and recognition program as well as the codes that communicate to the microcontroller the angles in which to position the servos that are tasked with keeping the tracking subject within the frame of the camera. The hardware involves the design and connections of the physical components that optimize the functions of the device while maintaining a clean and orderly appearance. The overall architecture of the neural network model can be seen in Figure 2.

The foundation of the recognition program can be traced back to the Mediapipe Hands[8] Library. The Mediapipe gesture recognition employed a bundle of four different deep-learning models. The first model detects the presence and location of the hand by focusing on palm recognition, the second model then locates the 21 landmark locations on the hand including joints and fingertips (as seen in Figure 3), the third model maps the detected landmarks into numerical values (vector embedding) and the final step of determining the hand gesture is carried out by the fourth model. The advantage of using Mediapipe is the convenience of having these underlying details carried out by the algorithm in a user-friendly format that receives images and outputs values.

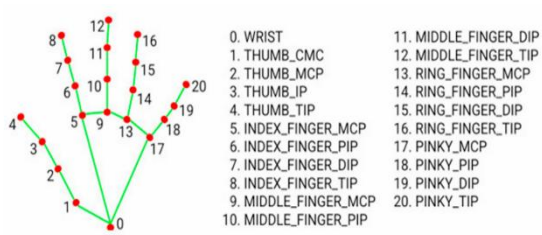

Figure 3: Hand Landmarks localized by Mediapipe Hands model

For a more detailed implementation of Mediapipe, the GitHub repository made by user Kazuhito00[9] was used as a

foundation to develop and train this project's hand recognition program. The repository included a sample program and two pre-trained recognition models, one recognizing simple hand signs and the other the moving gestures of the pointer finger, using a simple artificial neural network consisting of multiple layers of neurons, as well as the landmark detection function provided by Mediapipe Hands. The acquisition of the dataset for this project involves collecting hand landmarks from 4 subjects, 2 males and 2 females, each providing 20 iterations for each recorded hand sign in various lighting conditions and distances, 80 iterations per sign, totaling 80,640 values, not counting the irrelevant parameters. The training dataset was collected using the key point logging mode (Figure 4), capturing the individual coordinates of all 21 locations on the hand, then pre-processing these coordinates by normalization, converting pixel values into trainable data ranging from -1 to 1, as detailed in Table 1.

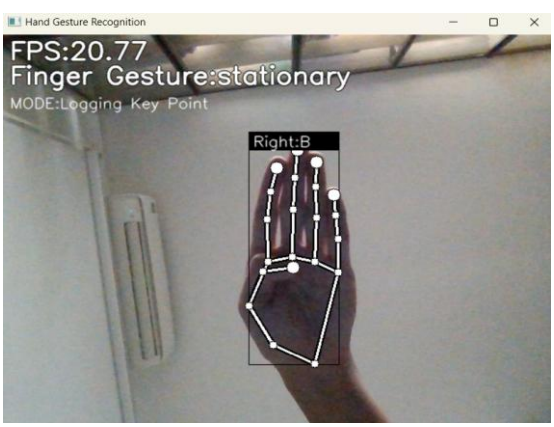

Figure 4: Video capture of hand recognition during data collection

Table 1. Process of data normalization

(1) Landmark coordinates					
ID: 0	ID: 1	ID: 2	.....	ID: 19	ID: 20
[551,465]	[485, 428]	[439,362]	.....	[687,225]	[702,188]
(2) Convert to relative coordinates from ID: 0					
ID: 0	ID: 1	ID: 2	.....	ID: 19	ID: 20
[0,0]	[-66,-37]	[-112,-103]	.....	[136,-240]	[151,-277]
(3) Flatten to a one-dimensional array					
ID: 0	ID: 1	ID: 2	.....	ID: 19	ID: 20
[0 0]	[-66 -27]	[-112 -103]	.....	[136-240]	[151 -277]
(4) Normalized to the absolute value					
ID: 0	ID: 1	ID: 2	.....	ID: 19	ID: 20
[0 0]	[-0.24 -0.13]	[-0.4 -0.37]	.....	[0.491 -0.87]	[0.545 -1]

For the training part of the process, Jupyter Notebook, or Ipython Notebook, was utilized. This tool offers a flexible environment for executing the training codes for data science and other machine-learning purposes. The sequential model is adopted for this project to create a Feedforward Neural Network with five layers, including two dropout layers and three dense layers. The summary of the model with the layer type, output shape, and parameters can be seen in Figure 5.

```
Model: "sequential_6"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 dropout_12 (Dropout)        (None, 42)                0

 dense_18 (Dense)            (None, 20)                860

 dropout_13 (Dropout)        (None, 20)                0

 dense_19 (Dense)            (None, 10)                210

 dense_20 (Dense)            (None, 24)                264

=================================================================
Total params: 1,334
Trainable params: 1,334
Non-trainable params: 0
_________________________________________________________________
```

Figure 5: The model summary of the neural network

A variety of epoch numbers were tested with the model training to see when the performance hits a plateau. The number of epochs that was settled on, which gives superior training results, was 400. The model will go through the total dataset 400

times, with each iteration processing 150 samples (batch size=150) before the model is updated, totaling a value of 60,000 batches for the entirety of this training process. After the training is completed and the classification yields the desirable results, the program can then be exported from the computer on which it was trained and imported into the Raspberry Pi. However, in order for the recognition to work within a largely different software environment of the Raspberry Pi 4, many adjustments need to be considered. Firstly, the Raspberry Pi computer does not have a built-in webcam, instead, in order to capture images for detection, a picamera module is required. After the module's installation onto the Pi, sections of the Python program for the model need to be altered or added to start using the picamera for hand recognition. The required libraries are then installed onto the Raspberry Pi using sudo commands (granting security privileges as the superuser) and pip3.

After all the libraries were installed, the model demonstrated accurate detection and recognition capability while running on the Pi 4. However, the framerates have reduced significantly from how it was performing on a PC due to the computational limitations of the Raspberry Pi. This being said, the integration of the model onto the small single-board computer for better portable applications was a success.

The servo control programs are then written directly on the pre-installed Thonny IDE of the Raspberry Pi and later integrated into the main hand detection codes. The servos are instructed to change their angles based on the return coordinates of a single landmark on the hand so that this specific landmark is always as centered on the frame captured by the camera as possible. Similarly, the codes to run the lighting unit are added. This works by allowing the Pi to receive information detected by the photoresistor and,

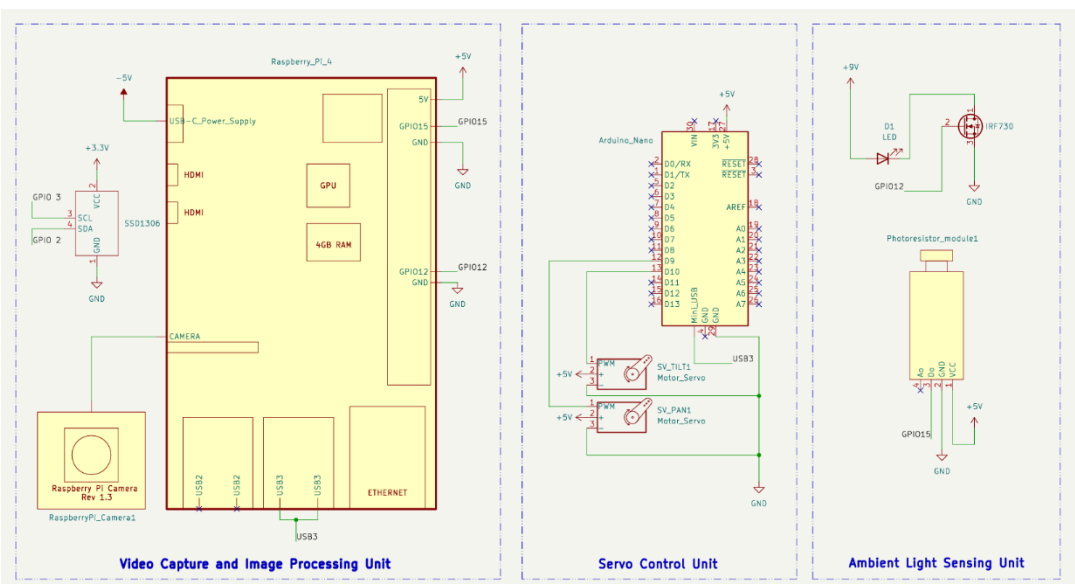

Figure 6: Schematic of the system

depending on whether there is enough light hitting the sensor, the LED attached to the unit is turned on or off, illuminating the detectable area. The hardware of this project is composed of a Raspberry Pi 4B, an Arduino Nano microcontroller, a photoresistor module, a Raspberry Pi camera module, two micro servos, a pan and tilt unit, and a module of 12 mini LEDs controlled with a MOSFET. Here, in the schematic, for simplicity purposes, a single typical LED. The wiring schematic of the entire system is illustrated in Figure 6.

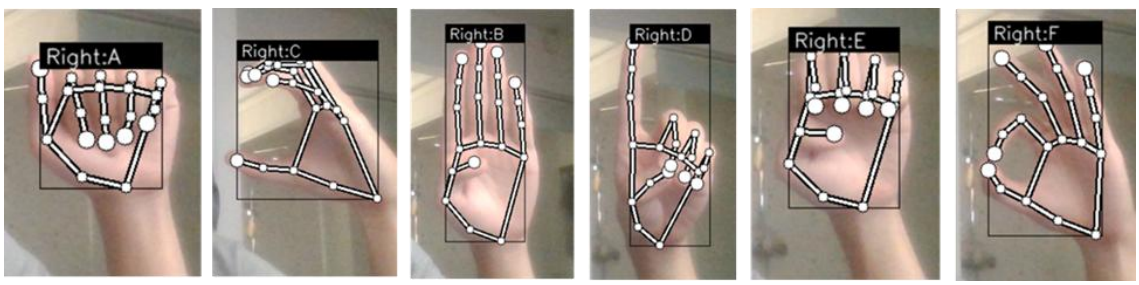

Figure 7: Detecting the first 6 letters of the alphabet

3. RESULTS AND DISCUSSION

In the experimental setup, once the hand assumes the appropriate position, the program promptly detects the corresponding sign language gesture. Subsequently, following a swift processing phase, the integrated system processes the quality of the ambient illumination surrounding the user and activates the lighting unit accordingly. The speaker component can be optionally triggered and emits a clear auditory output, effectively announcing the name of the interpreted letter as well as displaying the detected letter on the window monitor (Figure 7).

Concerning the accuracy of the hand recognition model, the program can accurately recognize a total of 24 letters in the English alphabet in ASL, excluding the two letters J and Z, which are not static and require moving gestures to express. The results of the practical tests are based on 30 tests per letter, with two male subjects and two female subjects, varying in the differences in hand sizes and lighting conditions. As expected, the Mediapipe-based model works effectively for all hand types and even for lower lighting conditions. The only circumstance where the model failed to detect the hands of the user is when the surrounding environment either has extremely low illumination levels or practically none at all, in which case the lighting module will be of service.

The accuracy and the loss of the model are evidenced by the graphs generated after training and testing, as seen in Figure 8. It can be observed that on the loss over epochs plot, the test loss is significantly lower than the training loss, which is uncommon. Given the nature of the model being trained on the training data, whereas the testing data is hidden, a more common and intuitive situation would involve the training loss being lower than the test loss. Similarly, the same can be seen in the model accuracy diagram, where the accuracy of the validation is remarkably better than that of the training set.

The reason for this is due to the dropout layers. During the training of the model, the dropout layers set a specific amount of features to zero, which creates slight fluctuations in the line graph. Meanwhile, in the validation process, all the neurons are used. This leads to higher testing accuracy due to the model used during testing being more robust. By using Sparse Categorial Cross Entropy, the model's loss was continuously reduced from about 3.5 to under 0.5 in 400 epochs.

Regardless of its atypicality, the results of the training and testing are still acceptable and sufficient for hand-recognition applications.

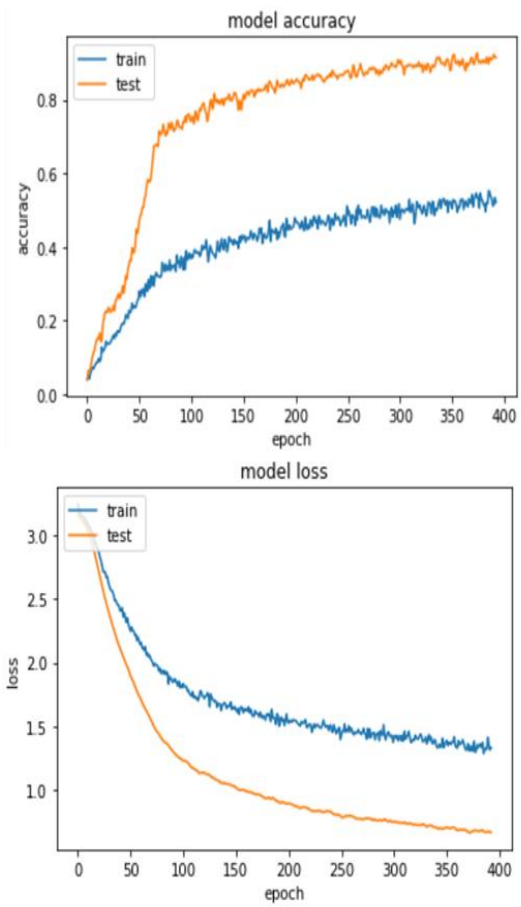

Figure 8: Accuracy and loss of the model

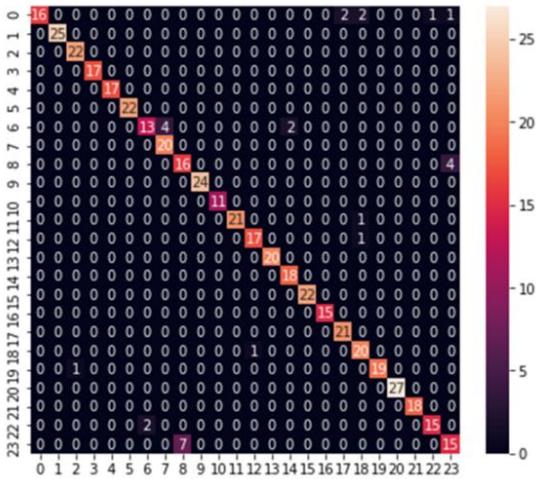

Figure 9: Confusion matrix of the model

The confusion matrix in Figure 9 demonstrates the accuracy of the model on the test data for 24 different hand postures. Overall, the accuracy reached 94%, with the lowest f1-score for individual letters being 0.71. This score belongs to the letter Y due to its similarity in posture with the letter I, which has an f1-score of 0.74. Both of the hand postures for these two letters share the extension of the little finger, with the only difference being the position of the thumb, which Mediapipe has difficulty distinguishing. Other than these cases, the rest of the signs from the ASL alphabet were detected and recognized with impressive accuracy, most having an f1-score of above 0.9.

However, several limitations were encountered during the development and testing phases of the project. Especially the exclusion of two of the letters from the alphabet, J and Z, due to their gestures demanding the finger to move across the screen, tracking the history of the detected landmark point of the pointer finger's tip, and training another model to perform

specifically such a task. This proves to be complicated and time-consuming, requiring a lot more effort than the hand posture recognition model to be efficient and work at an acceptable level.

Concerning the effectiveness of the sensor and hardware implementation's assistance for the hand recognition model. The lighting unit employing the photosensor works as anticipated, effectively providing illumination when required and assisting the hand recognition model. Similarly, the servo tracker application controlled by the microcontroller operated smoothly, following the user's hand once it had detected the presence of one, with minimal degrees of jitter. Finally, the total cost of fabrication is approximately $90, substantially less than most popular sign translator products on the market, fulfilling the project's requirement for affordability. The finished prototype of the device is depicted in Figure 10.

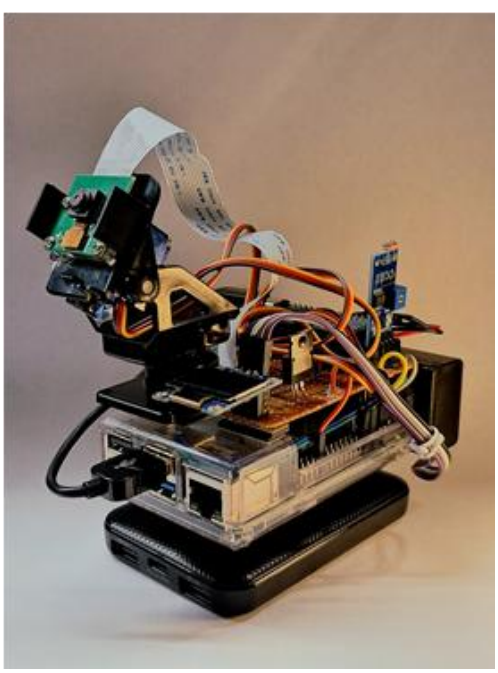

Figure 10: Prototype hand-sign translator

Another limitation is the inability to translate more comprehensive words, which in many cases require precise motion of the hand to express correctly. Future iterations can aim to overcome this limitation by incorporating higher-level translation capabilities. Furthermore, expanding language support beyond English to include other languages, such as Vietnamese, is a crucial area for future development.

4. CONCLUSION

All things considered, with hardware and budget limitations, the results yielded in this research are successful in translating 24 of the 26 letters in the ASL alphabet with some assistance from a system of different sensors and motors. Many more advanced technologies can be adopted into this design, creating an optimal device with increased accuracy, portability, and affordability. Further involvement of more state-of-the-art technologies in assistive technologies is under research and development.

Despite these challenges, the project accomplished its objectives by addressing and offering a viable solution for several criticisms made against the hand recognition method of employing vision-based technology and machine learning, mainly the issue with varying lighting conditions and self-occlusion affecting the accuracy of detection as well as the restricting space in which the hand of the user needs to be in for the program to recognize its positions and gestures. The offered answers to these problems were: (1) to employ a sensor that detects the

ambient lighting and activates additional built-in lights accordingly, (2) exploit the existing robust Mediapipe hand detection library when building the recognition model, and (3) install a pan-and-tilt unit for the camera unit so that once a hand is detected within the detecting frame, the unit will automatically rotate across its axes in order to accommodate the new position of the user's hand. The development of this project's sign language translator offers an idea that may be useful for future research in the field of assistive devices or image processing integrated technology in general. With additional recommended refinement and further development, there is enough potential in vision-based hand recognition technology to significantly enhance communication and exclusivity for many individuals worldwide.

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